

Hasnain Abbas

Skardu, Gilgit Baltistan, Pakistan

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Research Interests

Neuromorphic Computing and Memristive Devices Ferroelectric and Multiferroic Thin Films Sol-Gel Synthesis of Functional Ceramic Materials Resistive Switching and Synaptic Device Characterization Experimental Solid-State Physics

Education

Quaid-i-Azam University Islamabad

Islamabad, Pakistan

Master of Philosophy (MPhil) in Physics

Sep 2024 – Present (Expected Sep 2026)

- **Thesis:** Co-substituted BiFeO₃ Thin Films for Neuromorphic Computing and Memristor Applications
- **Supervisor:** Dr. Ghulam Hassnain Jaffari

COMSATS University Islamabad

Islamabad, Pakistan

Bachelor of Science (BS) in Physics, CGPA: 3.2/4.0 (79.9%)

Feb 2019 – Feb 2023

- **Major Coursework:** Solid State Physics, Statistical Physics, Quantum Optics, High Energy Physics, Biophysics
- **Thesis:** *Fast Forward Stimulated Raman Adiabatic Transfer of Atomic Population in Delta-Type Atomic Medium* (analytical modeling of accelerated atomic population transfer)

FG Boys Inter College Karachi Cantt

Karachi, Pakistan

Higher Secondary School Certificate (HSSC)

Apr 2016 – Aug 2018

Research Experience

Quaid-i-Azam University Islamabad

Islamabad, Pakistan

Graduate Researcher, Thin-Film and Memristor Lab

Sep 2024 – Present

- Synthesizing Co-substituted BiFeO₃ ceramic and polymer-blended thin films via the **sol-gel method**, optimizing precursor chemistry, annealing profiles, and film thickness for stable memristive switching
- Performing structural and compositional characterization using **XRD**, **FTIR**, thickness profilometry, and impedance analysis to correlate phase purity and microstructure with electrical response
- Measuring ferroelectric hysteresis (P-E loops) with a **Sawyer-Tower circuit ferroelectric tester** to quantify remanent polarization and coercive field across compositions
- Wrote custom **Lua automation scripts on a Keithley 2602 SourceMeter** to acquire high-resolution I-V data, enabling reproducible pulse and sweep protocols for memristive characterization
- Training **multilayer perceptron (MLP) neural networks in PyTorch** on the MNIST dataset, mapping experimentally extracted device conductance into the **Cross-Sim simulator** to evaluate inference accuracy of measured devices as synaptic weights

Pakistan Institute of Nuclear Sciences and Technology (PINSTECH)

Islamabad, Pakistan

Research Intern, Optics and LINAC Labs

- Trained across the Optics Lab and LINAC facilities and rotated through specialized high-energy physics labs
- Performed laser ablation of silver and gold nanoparticles for colloidal nanoparticle synthesis
- Generated and analyzed shock waves using a high-power laser system

Technical Skills

Fabrication	Sol-gel synthesis of ceramic and polymer thin films, spin coating, annealing, laser ablation (Ag and Au nanoparticles)
Characterization	XRD, FTIR, impedance analyzer, Sawyer–Tower ferroelectric tester, thickness profilometer, Keithley 2602 SourceMeter
Programming	Python (PyTorch, NumPy, Matplotlib), Lua (Keithley TSP), C++, Mathematica
Simulation & ML	Cross-Sim neuromorphic simulator, MLP training and evaluation on MNIST
Other Software	Adobe creative suite (image and video editing), L ^A T _E X
Languages	English (fluent), Urdu (fluent), Balti (native)

Teaching Experience

Gilgit Baltistan Institute of Science and Technology Skardu, Pakistan

Visiting Lecturer

Oct 2023 – Dec 2023

- Delivered lectures and supervised laboratory work for medical undergraduate students
- Demonstrated experiments and connected physics principles to medical sciences applications

FG Boys Degree College Skardu

Gilgit Baltistan, Pakistan

Visiting Lecturer

Mar 2023

- Taught BS Physics students and mentored them through coursework
- Supervised laboratory sessions for higher-secondary physics students

Conferences & Seminars

- 48th International Nathiagali Summer College on Physics and Contemporary Needs, Nathiagali, Pakistan (Jul 2023)
- Capacity Building Sessions for Lecturers, FG Boys Degree College Skardu, HEC Pakistan (Apr 2023)
- Workshop on Fundamentals of Quantum Mechanics, PIEAS, Islamabad (Jan 2023)

Awards & Honors

- **Best Lecturer Award**, Al-Zehra Girls College, Skardu (Dec 2023)
- **Best Organizer Award**, Seminar on Modern Physics, COMSATS University Islamabad (Aug 2020)

References

Dr. Ghulam Hassnain Jaffari

MPhil Supervisor

Department of Physics, Quaid-i-Azam University Islamabad

Email: hassnain@qau.edu.pk

Prof. Dr. Fazal Ghafoor

BS Supervisor

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Dr. Tasawar Abbas Malik

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Dr. Anwar Hussain

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